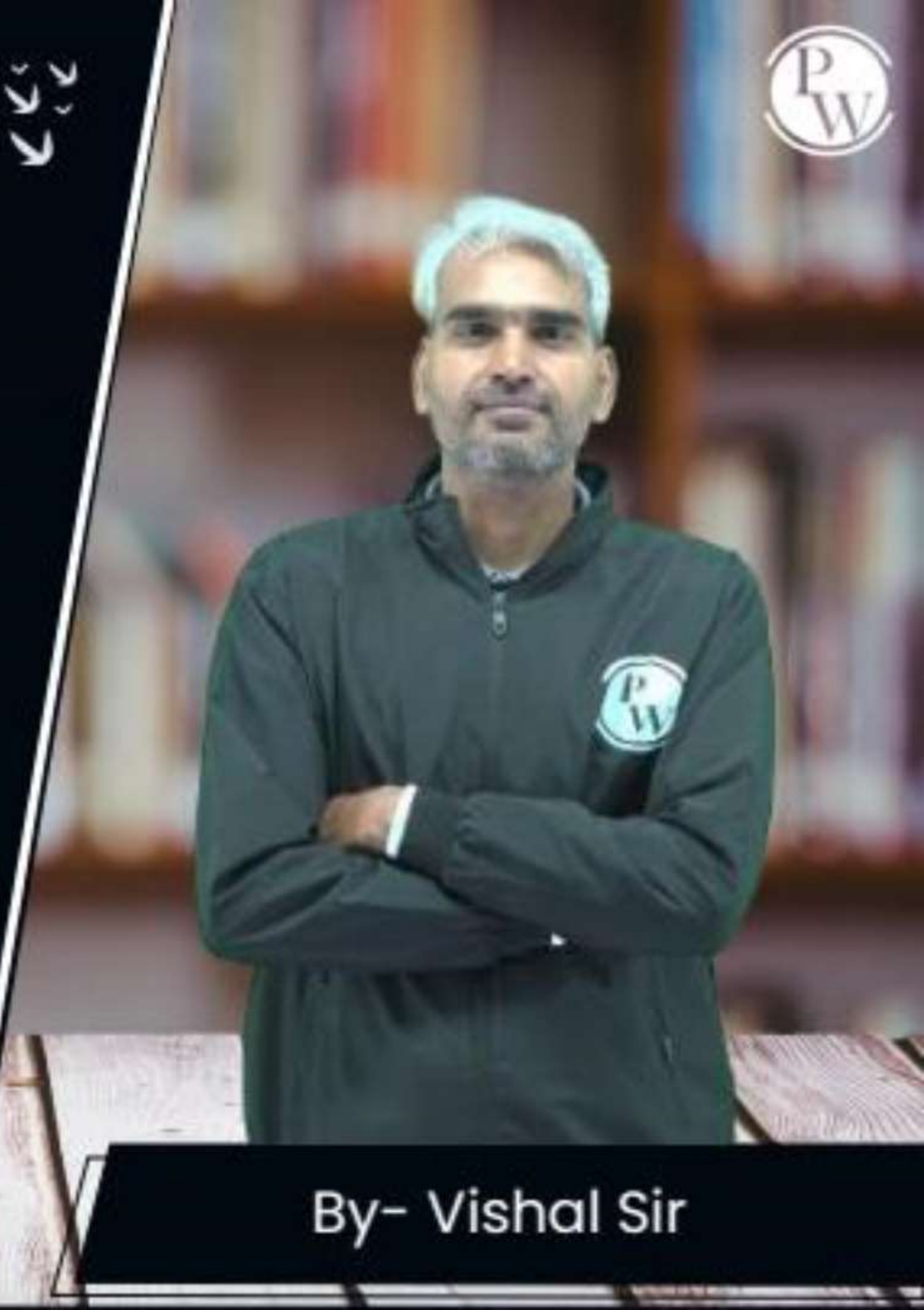
Computer Science & IT

Discrete Mathematics

Set Theory & Algebra

Lecture No. 12



By- Vishal Sir

Recap of Previous Lecture

- **Topic** Minimal / Maximal elements in a POSET
- **Topic** Minimum(Least) element of POSET
- **Topic** Maximum(Greatest) element of POSET
- **Topic** Lattice

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Topics to be Covered

Topic Hasse diagram / POSET diagram

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* **Note :→**

In a lattice least upper bound as well as greatest lower bound must exist for every Pair of elements and they must be Unique

It is constructed for a POSET

Topic : Hasse Diagram / POSET Diagram

In a Hasse diagram of a POSET, Reflexive, Anti-symmetric, Transitive

1. There is a vertex corresponding to every element of set.

i.e. (a,b) ∈ Relation

2. There is an edge from vertex a to vertex b if and only if a relates to b and there is no element x in the set such that a relates to x, and x relates to b.(Transitivity is implied in the Hasse diagram not represented explicitly)

$(a,x) \in R$ & $(x,b) \in R$

3. No self-loop on the vertices (i.e. reflexivity is implied in the Hasse diagram not represented explicitly).

4. It is not directed but it uses implied upward orientation.

i.e. $(a,b) \in R$ then in Hasse diagram b at upper side, a at lower side

w.r.t $(a,b) \in R$: a above b is wrong

Note: Some author define the Hasse diagram as directed graph.

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Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSETs

1) $(\{-1,0,2.5,4,6\}, \leq)$
2) $(D_6, \div)$
3) $(D_{12}, \div)$
4) $(\{2,3,4,6\}, \div)$
5) $(\{2,3,6,12\}, \div)$
6) $(\{1,2,3,4,6,9\}, \div)$

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(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$ less than or equal

Hasse diagram for given POSET.

$$\leq = \{(-1,-1), (-1,0), (-1,2.5), (-1,4), (-1,6),$$
$$(0,0), (0,2.5), (0,4), (0,6),$$
$$(2.5,2.5), (2.5,4), (2.5,6),$$
$$(4,4)\ (4,6)$$
$$(6,6)\}$$

$(-1,0) \in \leq$

$(-1, x) \in \leq$ & $(x, 0) \in \leq$

no such element in the set

$(-1, 2.5) \in \leq$
but we have '0' s.t.
$(-1,0) \in \leq$ & $(0, 2.5) \in \leq$
Hence, No direct edge b/w -1 & 2.5

(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$ less than or equal

Hasse diagram for given POSET.

- Given POSET is a TOSET as well.
- Whenever we construct the Hasse diagram for a TOSET, it will look like a linear Chain

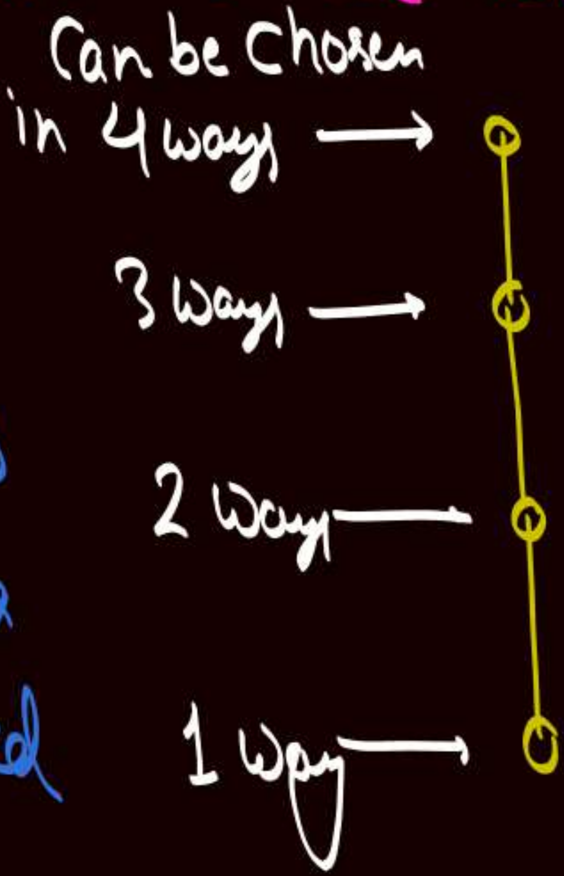
$\therefore$ Totally Ordered Set (TOSET) are also known as Linearly Ordered set.

Q:- Let $A = \{a, b, c, d\}$

How many Total order relation Can be defined on set A.

Hasse diagram for Every total order will look like a linear chain, but Hasse diagram of one total order is guaranteed to be different from other total order on that set.

$\therefore$ Number of total orders on a set with '4' elements will be Exactly same as the no. of ways in which different linear Chains of '4' elements Can be formulated



4 elements of the set Can be arranged at those '4' vertices in 4! ways

Note:- If A is a set with 'n' elements, then $n!$ different total order relation are possible on set A.

eg. Let $A = \{a, b, c, d\}$

let $R_1 = \{(a,a), (b,b), (c,c), (d,d), (a,b), (a,c), (a,d), (b,c), (b,d), (c,d)\}$ & $R_2 = \{(a,a), (b,b), (c,c), (d,d), (b,a), (b,c), (b,d), (c,a), (c,d), (d,a)\}$

Hasse diagram for (A, R_1) =

Hasse diagram for (A, R_2) =

Every Pair of element is Comparable w.r.t. R_1

$\therefore (A, R_1)$ is a TOSET

(A, R_2) is also a TOSET.

Hasse diagram will be different for different TOSET.

Note:- If POSET are different, then Hasse diagram will also be different on same set

eg. Let $A = \{a, b, c, d\}$

(i) $R_1 = \{(a,a), (b,b), (c,c), (d,d)\}$

└ it is partial order

∴ (A, R_1) is a POSET

Hasse diagram ≡

(ii) $R_2 = \{(a,a), (b,b), (c,c), (d,d), (a,b)\}$

└ it is partial order

∴ (A, R_2) is a POSET

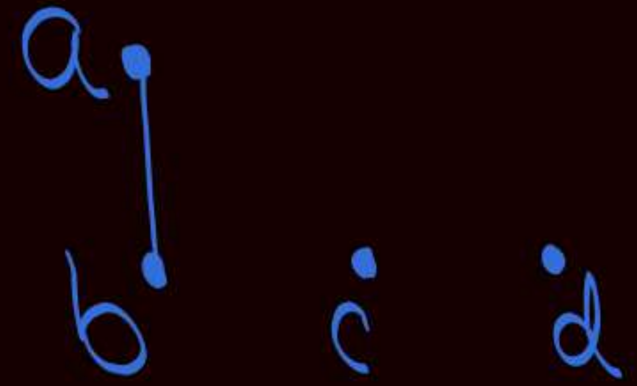
Hasse diagram ≡

(iii) $R_3 = \{(a,a), (b,b), (c,c), (d,d), (b,a)\}$

└ it is partial order

∴ (A, R_3) is a POSET

Hasse diagram ≡

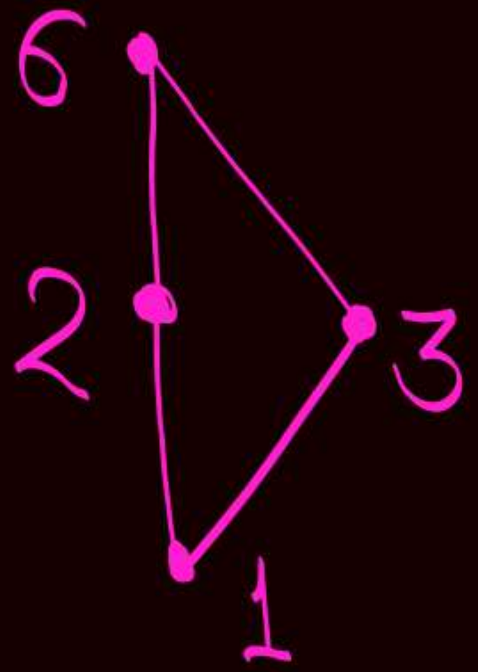


Q (ii) Construct the Hasse diagram for POSET,

$(D_6, \div)$

$D_6 = \{1, 2, 3, 6\}$

i.e, $(\{1, 2, 3, 6\}, \div)$



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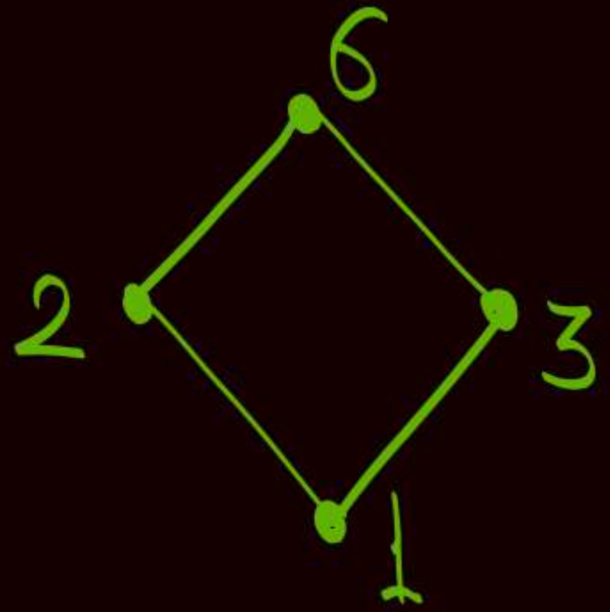
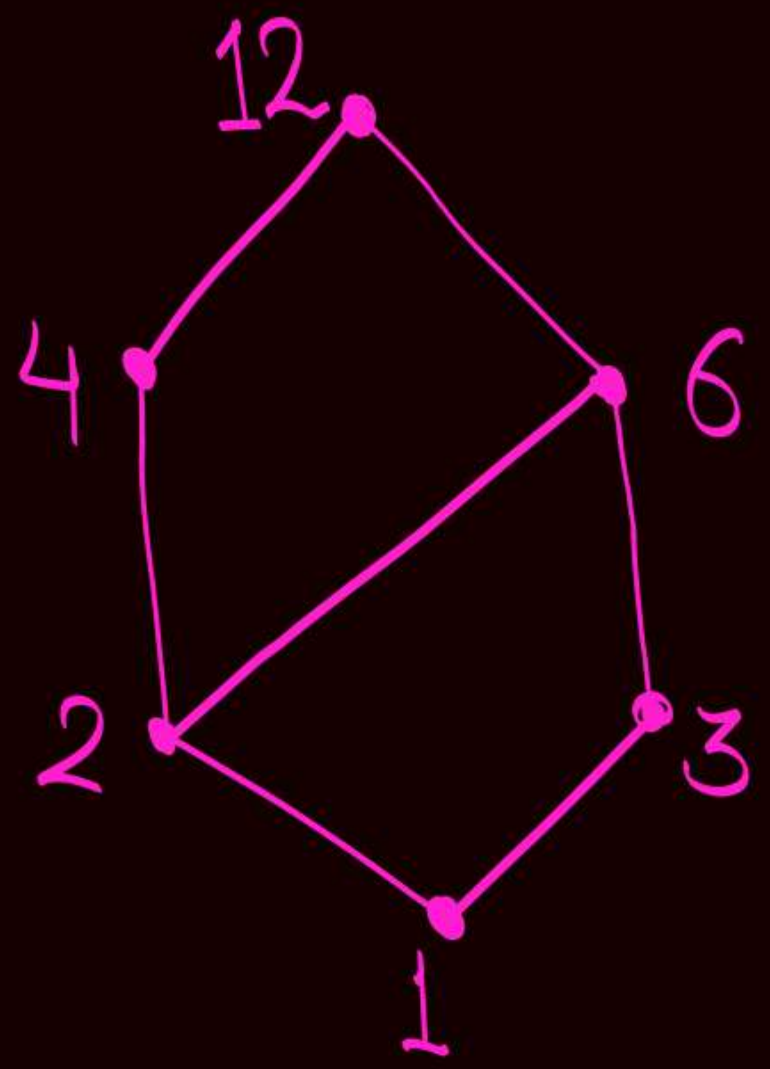


Q (iii) Construct the Hasse diagram for POSET,

$(D_{12}, \div)$

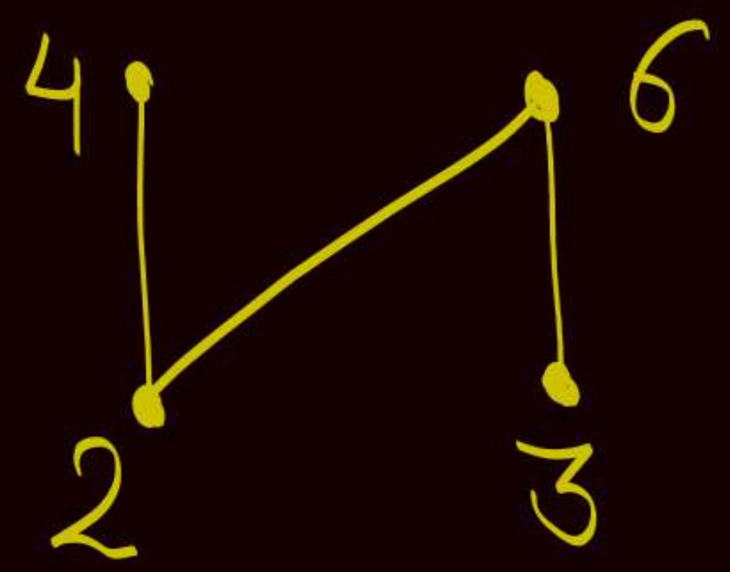
$\downarrow$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$



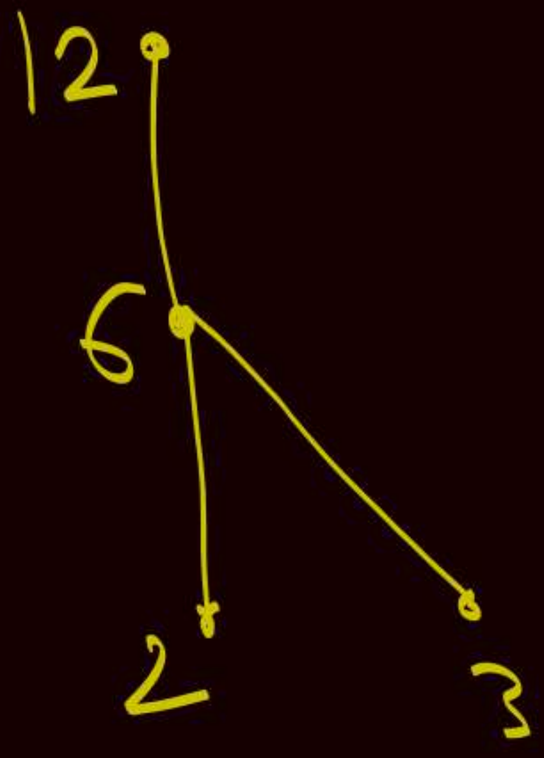
Q (iv) Construct the Hasse diagram for POSET,

$(\{2,3,4,6\}, \div)$



Q (V) Construct the Hasse diagram for POSET,

$(\{2, 3, 6, 12\}, \div)$



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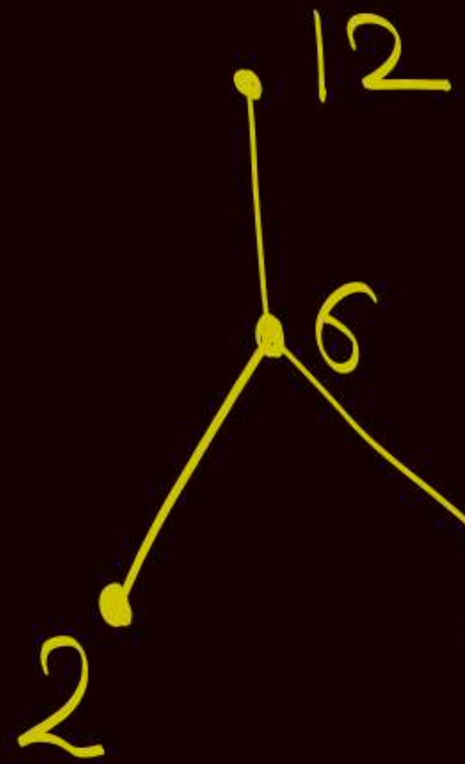
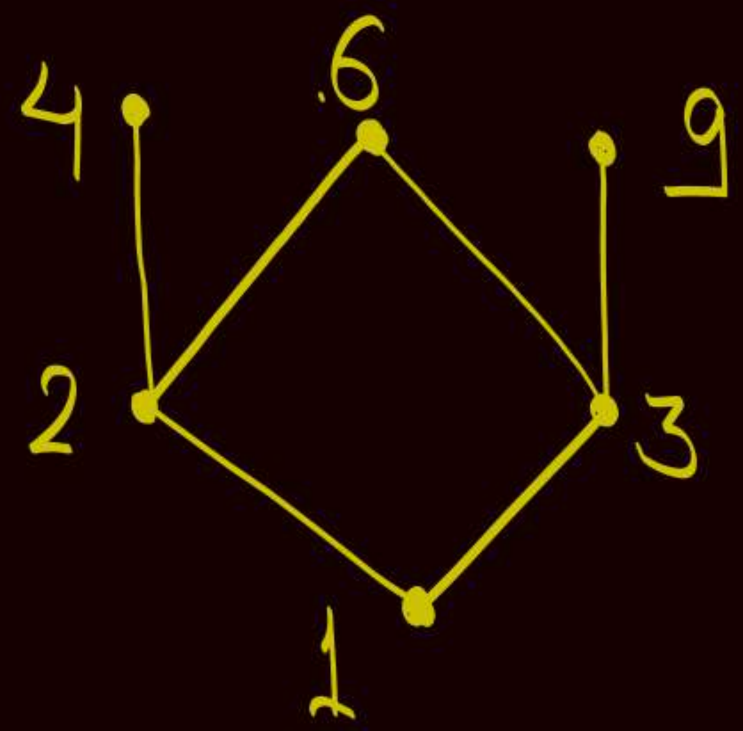


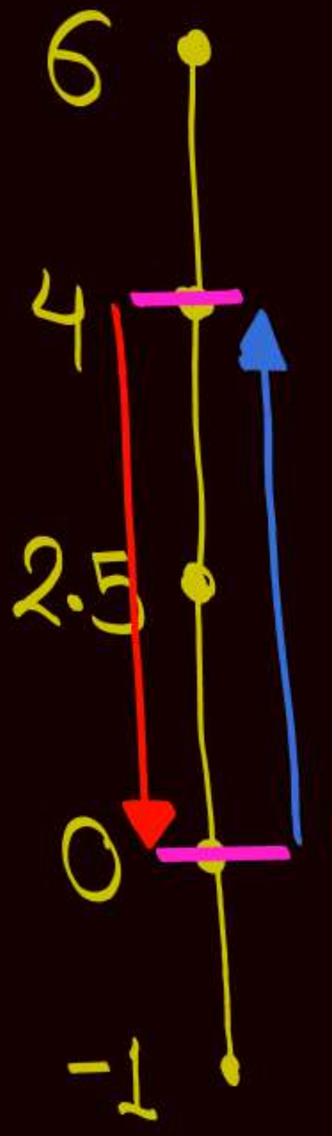
Q (vi) Construct the Hasse diagram for POSET,

$(\{1, 2, 3, 4, 6, 9\}, \div)$



Understand lub (Join) & glb (Meet) w.r.t. Hasse diagram

(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$



$lub(0,4) = 4$

$glb(0,4) = 0$

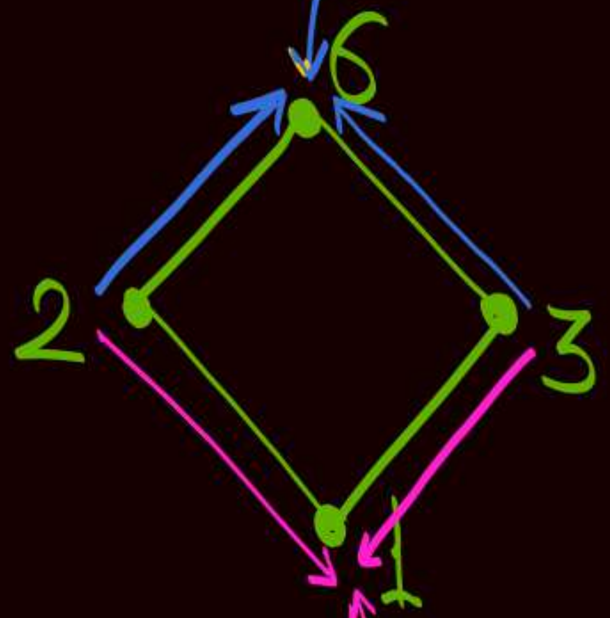
If we choose two elements on a line, then the element which is at the upper side is the lub, and the element at the lower side is the glb

In the given POSET both lub as well as glb exist for every pair of elements
$\therefore$ Given POSET is a lattice

Q (ii) Construct the Hasse diagram for POSET, $(D_6, \div)$

"2 & 3" join at '6'

$\therefore \text{lub}(2,3) = 6$



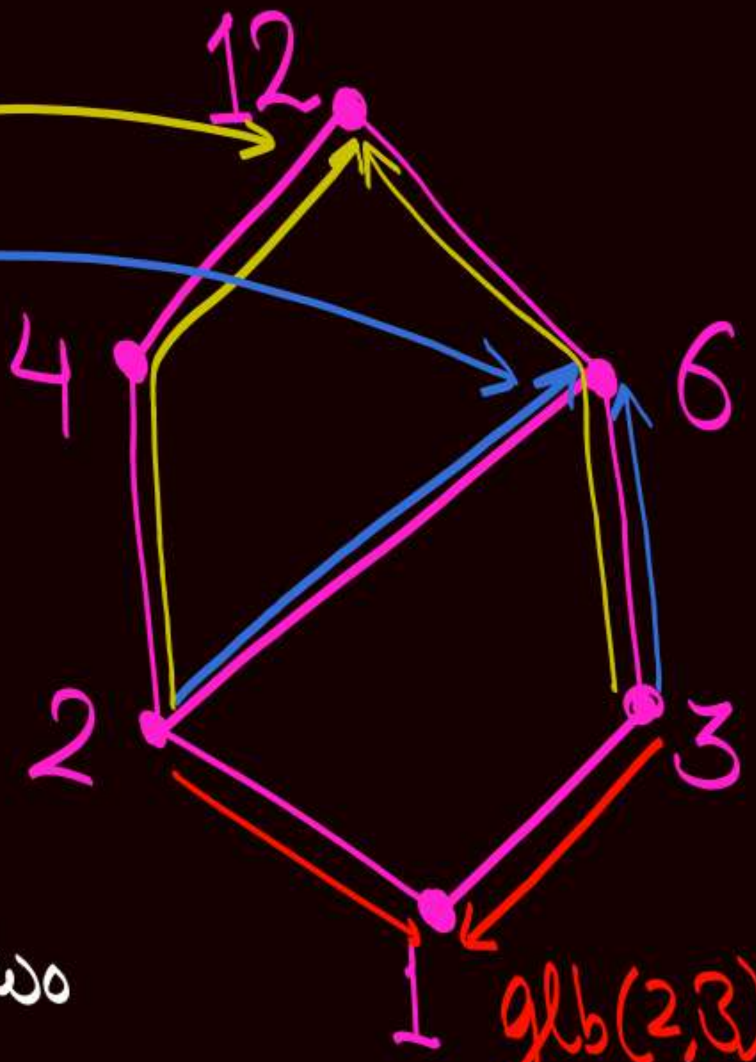
lub as well as glb exists for every pair of elements

$\therefore$ Given POSET is a lattice

"2 & 3" meet at '1'

$\therefore \text{glb}(2,3) = 1$

Q(iii) Construct the Hasse diagram for POSET, $(D_{12}, \div)$



$glb(2,3) = 1$

Both '6' & '12' are upper bounds of 2 & 3

but $(6,12) \in \div$

$\therefore$ 6 is least among two

Hence $lub(2,3) = 6$

$lub(2,3) = 6$

$glb(2,3) = 1$

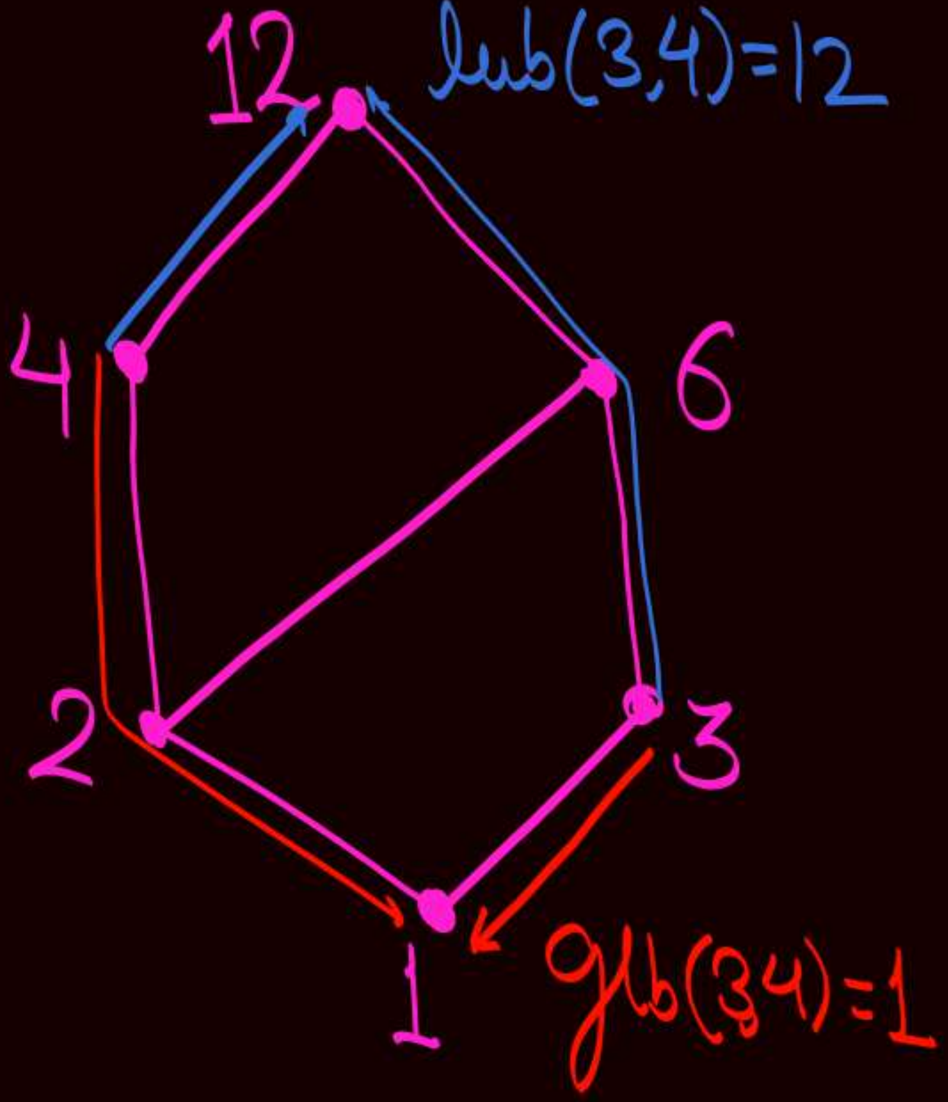
$lub(3,4)$

$glb(3,4)$

$lub(4,6)$

$glb(4,6)$

Q (iii) Construct the Hasse diagram for POSET, $(D_{12}, \div)$

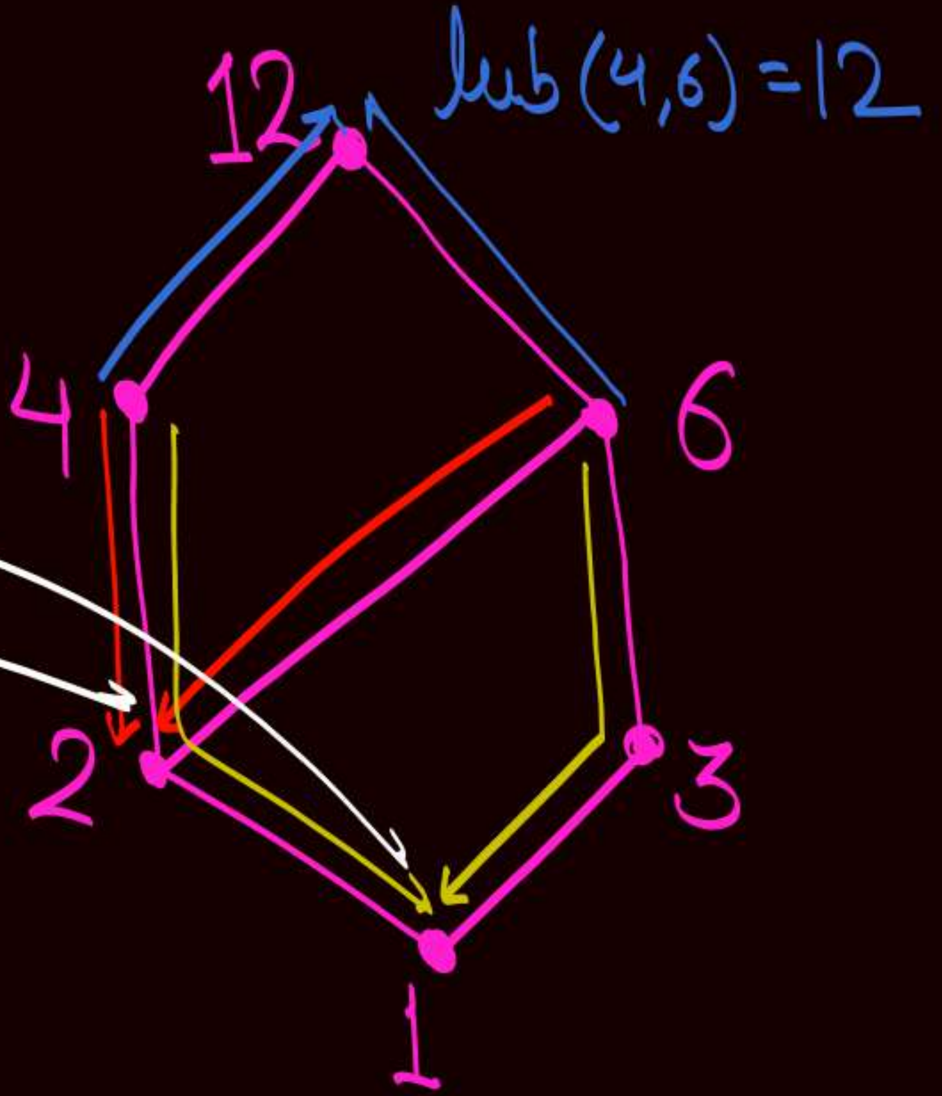


lub(2,3)
glb(2,3)

lub (3,4) = 12
glb (3,4) = 1

lub (4,6)
glb (4,6)

Q (iii) Construct the Hasse diagram for POSET, $(D_{12}, \div)$



Both 2 & 1 are lower bounds of 4 & 6

But greatest is '2'

$\therefore \text{glb}(4,6) = 2$

lub(2,3)

glb(2,3)

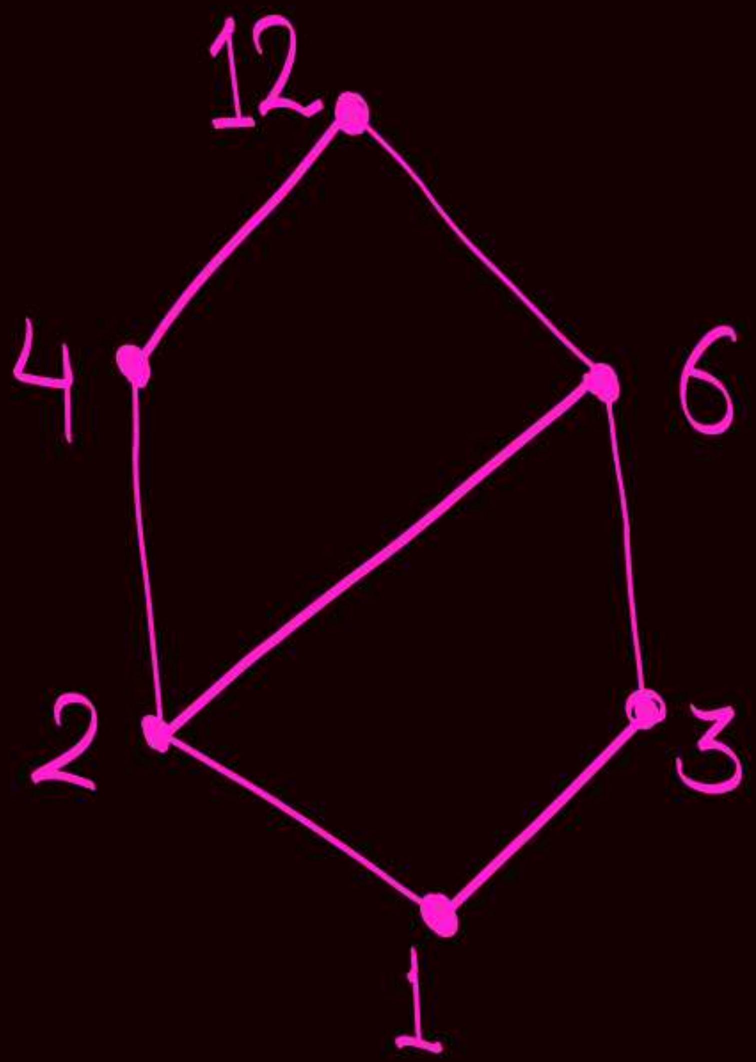
lub(3,4)

glb(3,4)

lub(4,6) = 12

glb(4,6) = 2

Q (iii) Construct the Hasse diagram for POSET,
$(D_{12}, \div)$



$lub(2,3) = 6$
$glb(2,3) = 1$

$lub(3,4) = 12$
$glb(3,4) = 1$

$lub(4,6) = 12$
$glb(4,6) = 2$

Lub as well as glb exists for Every pair of elements

$\boxed{\therefore \text{ Given POSET is a lattice}}$

2 mins Summary

Topic Hasse diagram / POSET diagram

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THANK - YOU

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